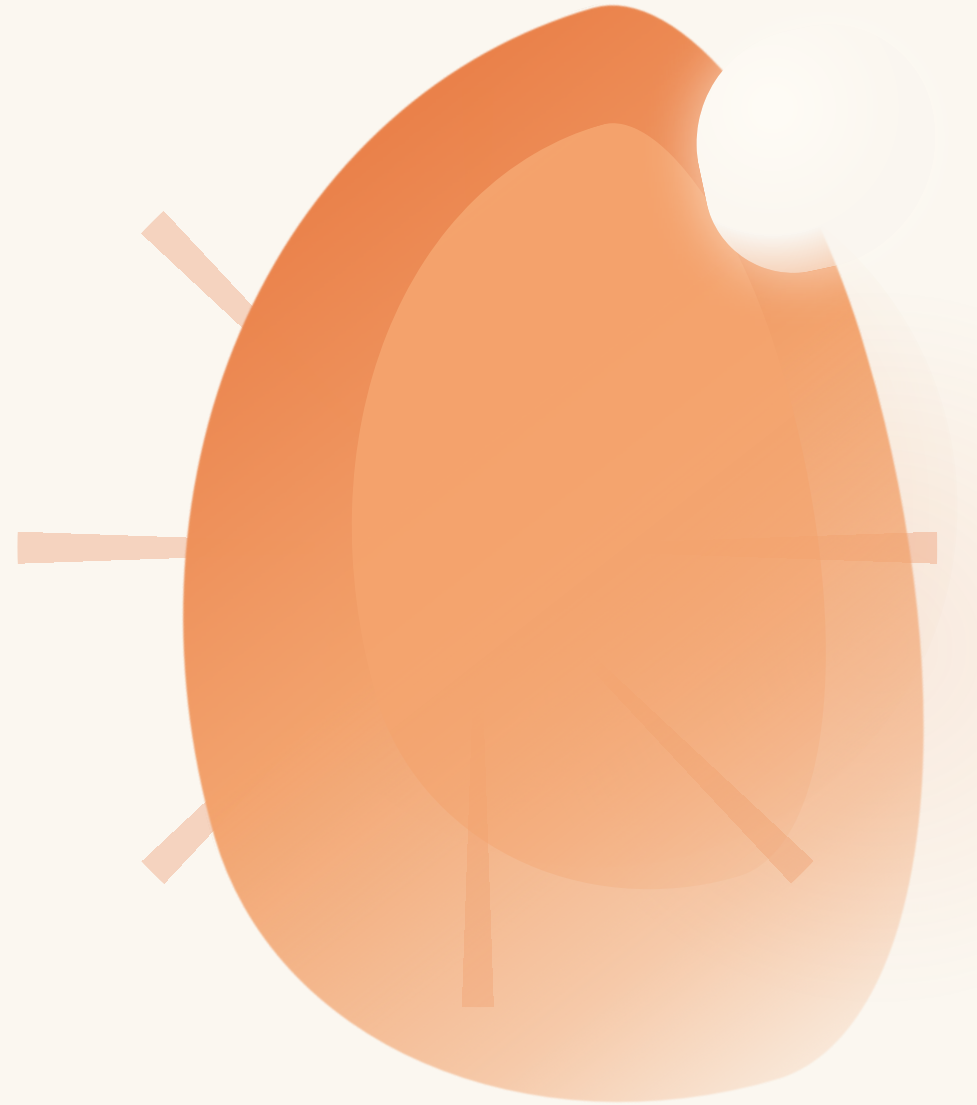

Suri

Offline AI Study Companion for Filipino Learners

"Matalinong kasama sa pag-aaral, kahit walang internet."

A study buddy that adapts to how you learn — offline or online, visual or auditory, standard or accessible.



Two compounding barriers Filipino students face

Existing EdTech platforms haven't resolved these together.

BARRIER 1 — CONNECTIVITY

28% of Philippine households had fixed home internet (2023)

35% of public schools have no internet connectivity at all

The device gap is mostly closed — almost all students already own a smartphone. The connectivity gap is not. Rural field studies confirm phone ownership is near-universal; **bandwidth is the bottleneck.**

BARRIER 2 — INCLUSIVE SUPPORT (RA 11650, 2022)

8% of an estimated 5M Filipino children with disabilities are enrolled in public schools (391,089 of ~5M)

60% of enrolled LWDs have no access to SPED programs, centers, or trained teachers

Only 12.3% of teachers hold SPED specialization. Classrooms of 40+ students with mixed needs and no aide are the norm. **The system cannot deliver on the Inclusive Education Act's mandate.**

Most EdTech platforms are built for students with consistent internet access and no learning differences. **Suri is built for the student who has neither.**

Three core capabilities, one companion

Suri is a React Native + Expo mobile app with a fox-mascot study buddy — built for the student who has neither connectivity nor learning-difference support.



Hybrid AI tutoring

16,000 RPD

free capacity — 80× the original single-provider design

A 3-tier cascade routes every request: **Gemini 3 Flash → Groq → OpenRouter** on good signal, reduced-payload cloud on weak signal, and a quantized on-device SLM (Qwen2.5-0.5B) when offline. Every response is grounded in local RAG of DepEd MELCs.



Adaptive learning modes

5 profiles

Visual · Auditory · Reading · Kinesthetic · Mixed

A Learning Profile shapes the format, depth, and modality of every AI response — across all three signal tiers. Visual learners get diagrams; auditory learners hear responses read aloud. **Same intelligence, different format.** Zero extra compute — it is a system prompt modifier.



Built-in accessibility

6 settings

Reader Font · Color Vision · High Contrast · Large Text · Focus Mode · Low Motion

Dyslexia-friendly font, colorblind-safe diagram palettes, ADHD focus mode, and low-vision support are user-selectable settings, not afterthoughts. None are labeled as disability modes — they are framed as **"how Suri talks to you."** Addresses RA 11650.

APK-first distribution

Shareable over local Wi-Fi, Bluetooth, or USB drives. No Play Store data charges. No Google account required. No bandwidth barrier at install time — schools can deploy without waiting for connectivity.

Always-on AI — three tiers, one experience

Tier transitions are automatic and invisible to the student. Provider failover within Tier 1 is also invisible.



Every response — every tier — retrieves **top-k relevant passages from a local RAG store of DepEd MELCs** before generation. Answers stay curriculum-accurate and grade-level-appropriate.

Why Suri is different

Same category. Different product. Built for the student who has neither.

DIMENSION	TYPICAL AI STUDY APP	✦ SURI
Offline mode	Caches pre-made content at best	✓ Full AI tutoring on-device via native llama.cpp — not just static content
Personalization	Same response format for every student	✓ Learning Profile adapts mode, modality, depth per student — zero extra compute
Accessibility	Not designed for learners with disabilities	✓ 6 comfort settings built-in — Reader Font, Color Vision, High Contrast, Large Text, Focus Mode, Low Motion
Language	English-first, Filipino bolted on via translation	✓ Multilingual cloud model primary; offline SLM selected for Filipino/Tagalog capability
Visuals	None, or fixed pick-from-library	✓ Generated fresh per question, grounded in MELC content, colorblind-safe by default
Distribution	Requires internet to install	✓ Shareable as APK over Wi-Fi, USB, or Bluetooth — no app store, no data charges
Data privacy	Cloud API by default	✓ Inference on-device by default — privacy by architecture, not policy

Suri is built for the student who has neither connectivity nor learning-difference support.

Live on stage — what we will demo

All demoable on a physical Android device. No special hardware. No pre-recorded video.



01

App shell

React Native / Expo. Offline detection.
Bottom-tab nav: Chat, Reviewer,
Quizzes, Profile.



02

Cloud AI streaming

Gemini 3 Flash via OpenAI-compatible
endpoint. SSE streaming. Provider
cascade shown by simulating rate-limit
hit.



03

Learning Profile setup

First-run quiz or direct selector. Visual /
Auditory / Reading / Mixed.
Personalization demonstrated
immediately.



04

Adaptive response — side-by-side

Same question, two profiles. Visual
returns a diagram. Auditory returns
clean prose + auto-plays TTS.



05

Working RAG

Pre-loaded Grade 6 Science MELCs in
SQLite. Grounded, grade-level-
accurate answer with source passage
cited.



06

Generative visuals

Bar chart + labeled science diagram
(e.g., plant cell) from LLM JSON spec
via react-native-svg.



07

Accessibility toggle — live

Reader Font (OpenDyslexic) + High
Contrast toggled on stage. Two taps,
instant effect.



08

Offline fallback

Kill the network. Cached SQLite
responses return instantly. If SLM pre-
loaded, show live local inference.

Plus: Suri mascot idle + thinking animated states. TTS Listen button on every response.

Validated — not hand-waved

Every mitigation below has been validated against confirmed library versions, API quotas, and device-compat matrices as of June 2026.

RISK · MODEL QUALITY

On-device SLM is weaker than the cloud model

Qwen2.5-0.5B cannot match Gemini 3 Flash on open-ended generation.

MITIGATION Model size reduced 62% (900MB → 350MB). Compensated at retrieval layer: top-k expanded 3 → 5 in offline mode. Output capped at 200 tokens. Response framed as structured summary, not open-ended generation.

✓ **VALIDATED** Qwen2.5-0.5B-Instruct-GGUF confirmed on Hugging Face; Q4_K_M compatible with llama.cpp

RISK · SINGLE POINT OF FAILURE

API provider dependency

Original design routed all cloud traffic through a single provider (OpenRouter, 200 RPD).

MITIGATION Three-provider cascade: Gemini 3 Flash (1,500 RPD) → Groq (~14,400 RPD) → OpenRouter DeepSeek V3 (200 RPD). Combined ≈16,000 RPD — 80× improvement. All OpenAI-compatible — single try/catch with three base URLs.

✓ **VALIDATED** Groq + Gemini OpenAI-compatible endpoints confirmed

RISK · QUOTA EXHAUSTION

Free API quota under classroom load

40 students × 40-minute period = 400–800 AI requests per class.

MITIGATION Cascade budget (≈16,000 RPD) comfortably covers this without touching the local SLM. SQLite response cache reduces repeat questions to cache hits. Rate-limit headers checked proactively — cascade switches providers before failing.

✓ **VALIDATED** Groq rate-limit response headers documented in official docs

RISK · DEVICE CONSTRAINTS

Lag and RAM pressure on budget Android

2GB RAM, 4-core CPUs, 2018–2022 devices — the most common device class in PH public schools.

MITIGATION 512-token context (saves ~400MB peak RAM). n_threads = min(4, CPU-1) — UI thread never starved. SLM pre-warmed 30s after launch. SQLite WAL mode. ML Kit + whisper.rn lazy-initialized on first use.

✓ **VALIDATED** cold ~8s → warm ~1s first-query latency

Android 8.0+ minimum covers ~97% of active devices globally — and the vast majority of student smartphones in Philippine schools manufactured 2018 onward.

What's next — and the one-line pitch

🚀 PHASE 2 — POST-HACKATHON ROADMAP

HIGH Code-switching support

Filipino/English/Tagalog code-mix. Matches how students actually speak. Formalized via prompt engineering + dialect detection.

HIGH Spaced-repetition scheduling

Turns the reviewer into a long-term retention tool. SQLite already tracks every answered question — the algorithm is the remaining work.

MEDIUM Parent / teacher dashboard

Sync-when-online progress visibility via Supabase backend. No requirement for the student to be always online.

LATER Classroom mesh sync

One connected phone passes a curriculum pack to a whole classroom over WebRTC / BLE.

🔊 THE ONE-LINE PITCH

“Suri is the only study companion that adapts to how you learn — visual, auditory, or accessible — working entirely offline on a Filipino student's phone, thinking in Filipino, and drawing a new diagram for every question instead of picking one off a shelf.”

🚩 What we ask from the judges

- 01 **Recognition** for offline-first, accessibility-first EdTech built for the Philippine context.
- 02 **Connections** to DepEd, DICT, or NGO partners for a pilot with 2–3 rural schools.
- 03 **Feedback** on the 3-tier cascade architecture and accessibility UX before pilot.